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Edge-Assisted Trust-Based Secure Communication in VANETs

23CSE399 – Project Phase 1 (Panel Review 2)

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1 Problem Definition & Originality

Problem Statement:

Vehicular Ad Hoc Networks (VANETs) enable real-time Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication and form a critical backbone of modern Intelligent Transportation Systems (ITS). These networks facilitate the exchange of safety-critical information such as accident alerts, emergency warnings, and traffic congestion notifications. However, despite the deployment of conventional security mechanisms like Public Key Infrastructure (PKI), VANETs remain highly vulnerable to insider attacks scenarios in which a legitimately authenticated vehicle deliberately behaves maliciously. Such vehicles can inject fabricated event data into the network, impersonate multiple identities through Sybil attacks, execute Denial-of-Service (DoS) flooding to suppress legitimate messages, or replay outdated alerts out of context. These attacks can cause severe real-world consequences including dangerous rerouting of emergency vehicles, suppression of genuine hazard warnings, and traffic accidents triggered by phantom obstacles. The fundamental problem is that existing authentication mechanisms verify *identity* but are entirely incapable of verifying *message truthfulness*, creating a critical and exploitable gap in VANET security that this project directly targets.

Originality Claim:

- **Extension:** This work extends existing VANET security by integrating Trust Management, Edge Computing, and Machine Learning into a single unified framework. Unlike prior works that combine at most two of these components, EdgeTrust-VANET treats all three as first-class architectural elements, enabling real-time, behavior-aware misbehavior detection at the network edge without centralized infrastructure.

2 Literature Survey & Gap Analysis

- **PKI and Authentication-Based Systems:** Standards such as IEEE 1609.2 and ETSI ITS-G5 rely on PKI for vehicle identity verification. While robust against external attackers, these mechanisms provide no defense against insider threats authenticated nodes are implicitly trusted regardless of message content or behavioral history.
- **Trust and Reputation Models:** Bayesian trust frameworks and watchdog-based reputation systems attempt to score node credibility based on historical interactions and peer observations. However, these models lack predictive capability, react slowly to sudden behavioral changes, and degrade in high-mobility environments where inter-node contact durations are too brief to build reliable trust histories.
- **ML-Based Intrusion Detection Systems:** Machine learning has been applied to VANET anomaly detection using Decision Trees, SVMs, and LSTMs. While promising in detection accuracy, nearly all proposed solutions adopt centralized architectures where raw data is aggregated at a remote server before classification, introducing latency incompatible with sub-second vehicular safety decisions.

- **Edge Computing in VANETs:** RSU-based edge processing has been studied to reduce communication latency. However, existing works treat RSUs purely as data relays or caching layers; their potential as intelligent, security-aware decision nodes integrated with trust evaluation and ML inference has not been exploited.

Research Gap:

- Absence of a unified framework combining Trust Management, ML classification, and Edge computing into a single operational pipeline for VANET security.
- Insufficient detection of insider misbehavior from legitimately authenticated nodes, which represent the most dangerous and prevalent threat class in deployed VANETs.
- No mechanism for real-time distributed misbehavior validation that scales with vehicular density without introducing a centralized performance bottleneck.
- Limited scalability in high-density VANET environments where thousands of nodes transmit simultaneously within overlapping RSU coverage zones.

3 Proposed Methodology (Technical Knowledge)

Core Concept:

EdgeTrust-VANET proposes a two-tier hybrid system combining dynamic Trust Score computation with Machine Learning classification, both deployed at Road Side Units (RSUs) acting as intelligent edge processing nodes. This ensures low-latency, real-time misbehavior detection without dependence on centralized infrastructure.

System Components:

- **Trust Module:** Computes a normalized per-node trust score $T \in [0, 1]$ at each evaluation cycle based on three criteria: (1) *Message Consistency* deviation of reported data from neighborhood consensus among nearby vehicles; (2) *Physical Plausibility* kinematic validation of reported speed, heading, and location against the vehicle's trajectory history; and (3) *Historical Behavior* a rolling behavioral profile that compounds past inconsistencies to enable proactive detection of slow-escalating attacks.
- **ML Classifier:** Trained on a feature set combining raw message attributes (speed, position, event type) and derived behavioral features (speed variance over a sliding window, heading deviation, message frequency anomaly score). The classifier outputs a maliciousness probability P_{mal} , which is fused with the trust score via a weighted decision function $S = \alpha(1 - T) + (1 - \alpha)P_{mal}$ to produce the final verdict.
- **Edge Layer (RSUs):** Each RSU operates as a fully autonomous security evaluation node. Upon receiving a vehicle message, it simultaneously computes the trust score and runs ML inference, issues an ACCEPT / WARN / BLOCK verdict within the evaluation cycle, and propagates BLOCK decisions to adjacent RSUs for network-wide isolation.
- **Fog Layer:** Provides higher-level coordination across RSUs. It maintains persistent longitudinal behavioral profiles, synchronizes trust state across RSU boundaries to handle vehicle handoffs, and aggregates decisions to support scalability under high vehicular density.

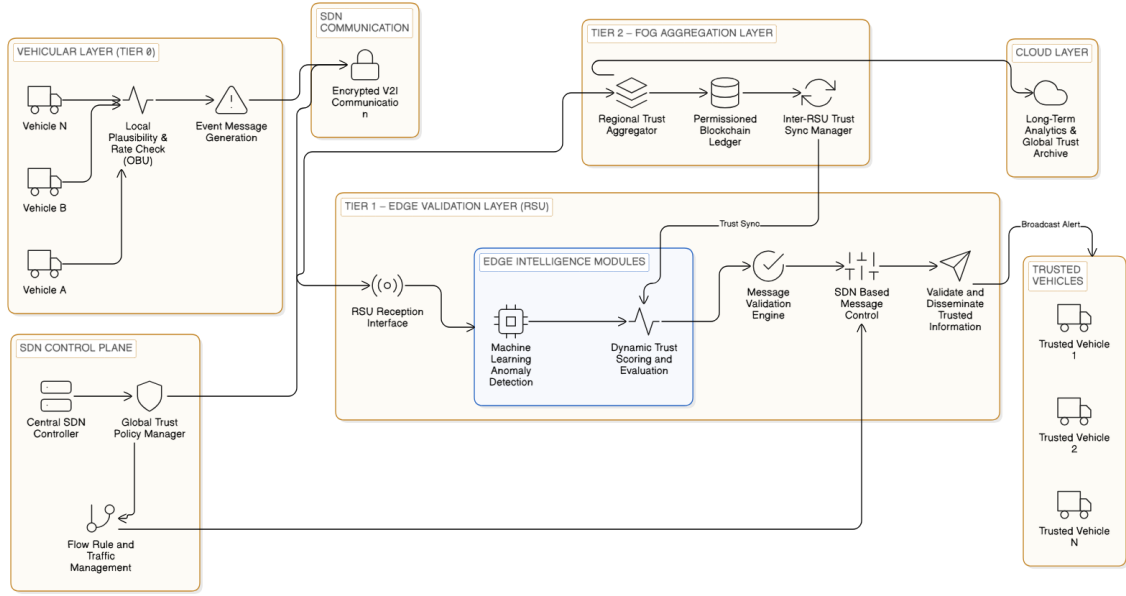


Figure 1: Proposed EdgeTrust-VANET Architecture

Technical Details:

- **Dataset:** 5000 vehicular nodes (25.2% malicious), with labeled instances spanning five attack categories: false data injection, Sybil, DoS, replay, and GPS spoofing.
- **Features:** Mobility features (speed, acceleration, heading deviation), Network features (message rate, neighbor count, signal strength anomaly), and Trust features (consistency score, plausibility score, behavioral deviation index).
- **Models Evaluated:** Random Forest, SVM (RBF kernel), and Gradient Boosting; all trained with 5-fold stratified cross-validation and compared on accuracy, precision, recall, and F1-score.
- **Best Model:** Random Forest achieved **95.6% accuracy** and **95.71% F1-score**, selected for deployment due to its robustness to noisy features, resistance to overfitting on tabular data, and fast inference time suitable for edge execution.

4 Implementation Details

Modules Developed:

- **Dataset Processing** (`analyze_vanet_nodes.py`): Loads raw vehicular data, removes leakage-prone and redundant features, applies min-max normalization, performs stratified train/test splitting, and outputs a clean feature matrix for model training and streaming inference.
- **Trust Computation** (`trust_score.py`): Implements the three-criteria trust scoring logic, outputs normalized per-node trust values at each evaluation cycle, and writes them to the fog layer's persistent behavioral store for longitudinal profiling across sessions.
- **Backend Server (Flask)**: Exposes RESTful APIs for triggering evaluation cycles (`/evaluate`), querying node status by vehicle ID (`/node/<id>`), and streaming live classification results (`/stream`) to the frontend dashboard.
- **Frontend Dashboard (HTML, JS)**: Visualizes real-time node trust scores, ACCEPT / WARN / BLOCK verdicts per vehicle, network-wide classification statistics, and historical behavioral trend charts updated dynamically via the backend stream.

Workflow:

- Vehicle transmits a message to the nearest RSU over the DSRC/C-V2X communication interface.
- RSU extracts the feature vector and simultaneously computes the trust score (`trust_score.py`) and runs ML inference via the loaded Random Forest model.
- Hybrid decision module fuses both outputs and generates the verdict: **ACCEPT** (trusted, message forwarded), **WARN** (borderline, flagged for monitoring with reduced routing weight), or **BLOCK** (malicious, message discarded and node isolated network-wide).
- Verdict and updated trust score are pushed to the fog layer; **BLOCK** verdicts are broadcast to all RSUs within the configured propagation radius for coordinated isolation.

5 Tools & Technologies

- **Python 3.10+, Scikit-learn:** Core ML pipeline for model training, hyperparameter tuning via grid search cross-validation, evaluation, and feature engineering.
- **Flask (Backend):** Lightweight REST API framework providing endpoints for real-time inference, node status queries, and inter-component communication between pipeline modules.
- **HTML5, CSS3, JavaScript (Frontend):** Responsive dashboard for live visualization of node trust scores, classification verdicts, and network-wide behavioral statistics.
- **SUMO, OMNeT++, Veins:** Traffic mobility simulation (SUMO) and network protocol simulation (OMNeT++ with Veins framework) for generating realistic vehicular trace data and validating system performance under controlled mobility scenarios.
- **Docker (Fog Layer):** Containerized deployment of the fog coordination service, ensuring reproducibility and portability of the trust synchronization layer across deployment environments.
- **Permissioned Blockchain ledger:** Planned for Phase 2 integration to enable tamper-proof, permissioned synchronization of trust scores across RSU administrative domains, preventing compromised edge nodes from manipulating the global trust state.

6 Current Status & Plan

Current Status:

- Literature survey completed across PKI-based authentication, trust and reputation models, ML-based intrusion detection systems, and edge computing architectures in VANETs.
- Dataset prepared and analyzed: 5000 nodes across five attack categories, feature engineering completed, stratified train/test split finalized with class balance verification.
- ML models trained and evaluated: Random Forest selected as primary classifier with 95.6% accuracy and 95.71% F1-score following cross-validated comparison against SVM and Gradient Boosting.
- Prototype dashboard implemented with real-time trust score visualization, live **ACCEPT** / **WARN** / **BLOCK** verdict display, and historical behavioral trend tracking per vehicle node.

Future Work:

- Integration with VANET simulators (SUMO + OMNeT++ + Veins) to validate detection performance under realistic vehicular density, mobility patterns, and RSU handoff scenarios across multiple coverage zones.
- Blockchain-based trust synchronization using Hyperledger Fabric to provide tamper-proof, decentralized trust state management, eliminating the single point of failure inherent in fog-layer coordination.
- Comprehensive performance benchmarking under active and adaptive attack scenarios including coordinated Sybil floods, replay campaigns, and adversaries trained to evade the detection pipeline; evaluation metrics to include latency, PDR, false positive rate, and throughput under varying node densities.

7 Conclusion

EdgeTrust-VANET presents a novel and practically motivated solution to one of the most critical unresolved challenges in vehicular network security the detection of misbehavior from legitimately authenticated insider nodes. By unifying dynamic trust score computation, machine learning classification, and edge-based processing at RSUs into a single cohesive framework, the proposed system moves decisively beyond the limitations of identity-only authentication mechanisms such as PKI. The Random Forest classifier, augmented with behavioral and trust-derived features, achieved an accuracy of **95.6%** and an F1-score of **95.71%** on a dataset of 5000 vehicular nodes spanning five attack categories, demonstrating robust and reliable detection performance across all evaluated threat scenarios. The edge-native deployment strategy ensures that classification verdicts are issued within the latency budget of real-time vehicular safety applications, making the system practically deployable in high-density ITS environments. With a clear Phase 2 roadmap encompassing VANET simulation integration, blockchain-based distributed trust management via Hyperledger Fabric, and adversarial robustness evaluation, EdgeTrust-VANET constitutes a meaningful and timely contribution toward securing next-generation intelligent transportation infrastructure.